



Assessment Brief

Module title:	Maths for Data Science
Module code:	COM7023
Assignment title:	Maths for Data Science Portfolio
Assignment format:	Report
Word/time limit:	3000
File type	.docx or PDF file
Percentage of final grade	This assignment is worth 100% of your final grade for this module.
Submission deadline	See module iLearn page for date of submission
Grade release	You will normally receive your provisional grade and feedback within 20 working days of the submission deadline

Useful terms:

Learning outcomes (LOs)	The skills and knowledge that you should be able to show in your work.
Rubric/Marking Matrix	A set of rules or guidelines used to grade or assess work.

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Task summary:

This assessment requires the submission of a written Mathematics for Data Science assignment. It evaluates your advanced understanding and application of key mathematical concepts underpinning data science, including Linear Algebra, Calculus, Probability, and Statistics.

You must demonstrate both theoretical knowledge and practical application within a data science context. All tasks should be presented in a single Word document, or PDF file, supported by screenshots of computational processes and results. Mathematical notation must be produced using appropriate software-based tools, and handwritten work is not permitted. All data analysis and visualisation must be carried out using Python, and your student ID must be clearly stated on the submission.

Failure to meet these requirements may result in penalties in line with programme regulations.

Assignment instructions:

You have recently joined the **Urban Mobility Intelligence Unit (UMIU)**, a consultancy supporting the UK Department for Transport (DfT) in improving the safety, efficiency, and performance of transport systems across major UK cities. As part of your induction, your Line Manager has asked you to demonstrate your capability in applying advanced mathematical techniques to a real-world smart mobility analytics problem before progressing to more complex projects.

The investigation is situated within the context of smart mobility operations in Manchester, where initiatives such as adaptive traffic signal control, electric vehicle (EV) charging corridors, autonomous vehicle trials, and Intelligent Speed Assistance (ISA) technologies are being deployed.

Your work will involve applying mathematical concepts from Linear Algebra, Calculus, Probability, and Statistics to analyse different aspects of urban traffic behaviour and system performance. In particular, the investigation aims to support the agency in understanding how EV charging infrastructure affects equilibrium traffic flows, how vehicle speeds and acceleration patterns vary along smart corridors, how uncertainty in autonomous vehicle reaction times can be quantified, and whether observed vehicle speeds comply with regulatory limits in ISA-equipped zones.

The initial part of the investigation focuses on modelling steady-state traffic flow across a network of interconnected junctions affected by EV charging activity. The system is represented by the following set of equations:

$$2x_1 - x_2 + x_3 = 150$$

$$x_1 + x_2 + x_4 = 420$$

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$$\begin{aligned} -x_1 + 3x_3 - x_4 &= 100 \\ x_2 + x_3 + x_4 &= 380 \end{aligned}$$

Where:

x_1, x_2, x_3, x_4 represent traffic flows (vehicles per 10 minutes) along four key roads. You are expected to analyse this system using appropriate linear algebra techniques, determine whether a stable equilibrium exists, and interpret the resulting traffic distribution in the context of EV-related stopping and merging behaviour.

The investigation then considers vehicle speed behaviour along a smart mobility corridor where adaptive traffic signals introduce periodic variations in speed. Vehicle velocity at time t seconds is modelled by:

$$v(t) = 70 + 12 \cos\left(\frac{\pi t}{5}\right) - 4 \sin\left(\frac{2\pi t}{5}\right) \text{ (km/h)}$$

You should analyse the corresponding acceleration behaviour and evaluate periods of rapid acceleration and deceleration. Your discussion should consider the operational and safety implications of these patterns, particularly in relation to braking behaviour, passenger comfort, and corridor optimisation.

To support autonomous vehicle safety assessment, the investigation further examines uncertainty in emergency-braking reaction times. Reaction times are assumed to follow a normal distribution:

$$R \sim N(\mu = 0.85 \text{ s}, \sigma = 0.25 \text{ s})$$

You are expected to analyse this distribution to evaluate the likelihood of rapid and delayed reactions and to interpret how probabilistic measures, such as percentile values, inform safety certification and system design.

Finally, the investigation evaluates vehicle speed compliance within an ISA trial zone with a legal speed limit of 50 km/h. The following sample of observed vehicle speeds (km/h) has been recorded:

$$49, 52, 55, 48, 50, 53, 57, 51, 54, 49, 52, 56$$

You should apply appropriate statistical techniques to analyse this data and evaluate whether the observed driving behaviour is consistent with regulatory expectations. The implications of your findings for ISA effectiveness and transport policy should be clearly discussed.

Once all numerical analyses have been completed, you must **critically evaluate** the mathematical approaches used across the investigation. Your final report should reflect on assumptions, limitations, and strengths of the models applied, and consider how effectively the analysis supports decision-making in smart urban mobility systems. Where appropriate, you may suggest improvements, alternative approaches, or creative extensions.

The completed report must be written for non-technical stakeholders within the Urban Mobility Intelligence Unit and should focus on the meaning and implications of the results, rather than procedural detail.

(100 Total marks)

(2000 words equivalent)

(1000 words for written results)

(LO's: 1,2,3 & 4)

Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on **all** four of the learning outcomes:

LO1. Develop an advanced understanding of mathematical concepts and their application in data science.

LO2. Apply mathematical principles to solve data science-related problems.

LO3. Deploy mathematical functions, models, and equations to derive advanced analytical solutions.

LO4. Graduate Attribute. Identify and solve novel and complex problems related to aims and desired outcomes. Critically evaluate and reflect on the approaches and solutions identifying and embedding possibilities for originality or creativity.

Your marker will use a rubric/ marking matrix to grade your work, and you can access this by clicking on the submission portal.

Guidelines and policies

You can find links to more useful information about the assignment and university policies below.

Word/time limit policy

[Click here to view the Arden University word count/time limit policy](#)

Referencing guidelines

[Click here for Harvard referencing guidelines](#)

Please follow the referencing guidelines that are appropriate for your degree programme. If you are unsure which you should be using, please contact your module team.

Academic integrity and misconduct policy

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